

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0606 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 20%

QUESTIONS: 05 Total Marks:100 Annexure A

Industry Problem Solving

Code of Conduct:

*I Prudhvi Raj.B(192324274) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:prudhvi raj.b

Annexure A

1. A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.
2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.
3. A healthcare company aims to classify patients into risk categories using historical medical data. Suggest a model using KNN or RBF networks. Explain how the algorithm handles large datasets, feature selection, and accuracy improvement in real-world scenarios.
4. An e-commerce platform wants to detect similar products for dynamic pricing strategies. Design a solution using Case-Based Learning. Discuss how similarity metrics are defined, how past cases are reused, and how the system scales with increasing product data.
5. A logistics company needs to forecast delivery times based on location and traffic patterns. Propose a solution using Radial Basis Function networks or Locally Weighted Regression. Explain how the model captures non-linear relationships and improves prediction reliability under varying conditions.

RUBRICS FOR CALCULATION :

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Retail Product Recommendation – KNN

Problem: Recommend products based on customers' previous purchases.

Solution: Use K-Nearest Neighbour (KNN).

- *Represent each customer using features such as products purchased, purchase frequency, spending amount, and product categories.*
- *Calculate similarity using Euclidean distance or Cosine similarity.*
- *Find the K customers most similar to the target customer.*
- *Recommend products purchased by those similar customers but not yet purchased by the target customer.*
- *When new purchase data arrives, the system can include it in the customer database and update recommendations.*

Example:

If Customer A has similar purchasing behaviour to Customers B, C, and D, products frequently purchased by B, C, and D can be recommended to A.

2. Manufacturing Machine Output – Locally Weighted Regression

Problem: Predict machine output from sensor readings that change over time.

Solution: Use Locally Weighted Regression (LWR).

- *Collect sensor readings such as temperature, pressure, vibration, and speed.*
- *Give higher weights to observations closer to the current query point.*

- *Build a local regression model around the query.*
- *Predict machine output using this local model.*
- *Unlike ordinary linear regression, LWR can capture changing relationships over time.*

Handling noisy/incomplete data

- *Remove or replace missing values using imputation.*
- *Use smoothing or outlier detection to reduce sensor noise.*
- *Adjust the bandwidth parameter to control how many nearby observations influence the prediction.*

Advantage: The model focuses on the most relevant recent/similar observations rather than treating every observation equally.

3. Healthcare Patient Risk Classification – KNN

Problem: Classify patients into risk categories using historical medical data.

Solution: Use K-Nearest Neighbour (KNN).

Possible features:

- *Age*
- *Blood pressure*
- *Cholesterol*
- *Blood sugar*
- *Heart rate*
- *BMI*

The system finds patients with similar medical measurements and assigns the new patient to the

majority risk category among the nearest neighbours.

For large datasets

- *Use feature scaling because KNN depends on distance.*
- *Remove irrelevant features.*
- *Use dimensionality reduction such as PCA if necessary.*
- *Use an efficient neighbour-search method for large datasets.*

Accuracy improvement

- *Select relevant medical features.*
- *Test different values of K.*
- *Use cross-validation.*
- *Handle missing values and outliers.*
- *Balance classes if one risk category has very few patients.*

Example:

New Patient



Calculate similarity



Find K similar patients



Majority voting



Low / Medium / High Risk

4. E-Commerce Similar Product Detection – Case-Based Learning

Problem: Find products similar to a given product for dynamic pricing.

Solution: Use Case-Based Learning (CBL).

Each previous product is stored as a case containing:

- ***Product category***
- ***Brand***
- ***Price***
- ***Rating***
- ***Features***
- ***Demand***
- ***Sales history***

Similarity

Similarity can be calculated using:

- ***Euclidean distance for numerical features.***
- ***Cosine similarity for product descriptions/features.***
- ***Weighted similarity when some features are more important.***

Case reuse

When a new product arrives:

New Product



Compare with previous products



Find most similar products



Analyze their prices and demand



Suggest suitable price

The system can continuously add new products and pricing outcomes to its case database.

Scaling

For a large number of products:

- *Use databases/indexing.*
- *Use feature vectors.*
- *Use approximate nearest-neighbour search.*
- *Remove outdated or duplicate cases.*

5. Logistics Delivery-Time Prediction – RBF Network

Problem: Predict delivery time based on location and traffic conditions.

Solution: Use a Radial Basis Function (RBF) Network.

Possible inputs:

- *Distance*
- *Starting location*
- *Destination*

- *Traffic level*
- *Time of day*
- *Weather*
- *Vehicle type*
- *Historical delivery time*

The RBF network uses Gaussian functions to measure how close a new delivery situation is to learned patterns.

Location + Traffic + Distance + Time



RBF Layer



Non-linear Patterns



Output Layer



Predicted Delivery Time

Why RBF?

RBF networks can model non-linear relationships.

For example:

- *10 km with low traffic → 20 minutes*
- *10 km with heavy traffic → 50 minutes*

The relationship between distance, traffic, and delivery time is not necessarily linear.

Improving reliability

- *Train using historical delivery data.*
- *Include different traffic and weather conditions.*
- *Remove abnormal delivery records.*
- *Normalize input features.*
- *Regularly retrain the model with new delivery data.*

Result: The system can provide more reliable delivery-time estimates under changing traffic and location conditions.